



Oromia National Regional State

Integrated Land Service System (ILS)

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Technical Project Proposal

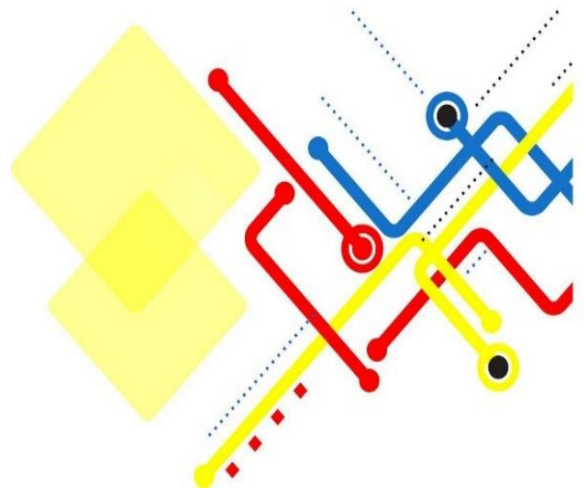
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I. Revision History

Date	Version	Description	Author	Reviewer
Oct 2022	1.0	Project Proposal Document for Oromia National Regional State Land Office Integrated Land Service System development.	Tilahun Dessalegn	Dawit Abel Mesfin Tafese

II. Approvals

This document requires the following approvals and the signed copy should be placed in the project files.

Name	Signature	Title	Date of Issue	Version
Miss. Tigist Hamid (INSA)		Deputy Director	Oct 2022	1.0
Ato Tilahun Dessalegn (INSA)		Cadaster System Project Division Head	Oct 2022	1.0

III. Distribution

This document has been distributed to:

Organization	Date of Issue	Role	Version
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IV. Acronyms

INSA	Information Network Security Administration
ILS	Integrated Land Service
SRS	System Requirement Specification
SAD	System Architecture Document
ONRSLB	Oromia National Regional State Land Bureau

QA	Quality Assurance
UAT	User Acceptance Testing
TOT	Training Of Trainers
ETB	Ethiopia Birr

Table of Contents

I.	Revision History	i
II.	Approvals	i
III.	Distribution	i
IV.	Acronyms	i
	List of Table	v
	List of figure.....	vi
1.	Introduction.....	1
1.1.	Background	1
1.2.	Document purpose.....	3
1.3.	Audience.....	3
1.4.	Objectives.....	3
1.5.	Project Scope.....	4
1.5.1.	In Scopes.....	4
1.5.2.	Out Scopes.....	5
1.6.	Project Assumptions	5
1.7.	Project Constraints	6
1.8.	Project tolerances	6
2.	Project Impact	7
2.	Overview of the project (ILS).....	8
9.1.	High level requirements.....	8
9.2.	High level system architecture.....	12
10.	Project stakeholders	13
11.	Project Approach.....	13
12.	Requirement Analysis	14
13.	System architecture design.....	16
14.	Software QA/QC/test plan	19
14.1.	System landscape.....	19
14.1.1.	General overview	19
14.1.2.	Sizing of each platform	20
14.2.	Source code quality.....	20
14.2.1.	Code documentation.....	20
14.2.2.	Programming style	20
14.2.3.	Code quality	21
14.3.	Software architecture	21
14.4.	Development platform	22

14.4.1.	General overview	22
14.4.2.	Development platform software	22
14.4.3.	User types and roles	23
14.4.4.	User registration	23
14.4.5.	Source code hosting.....	23
14.4.6.	Continuous integration system	24
14.4.7.	Continuous code inspection system	24
14.5.	Test platform.....	24
14.5.1.	Test platform software.....	25
14.5.2.	Performance testing.....	25
14.5.3.	Functional testing	25
14.5.4.	Bug tracker	25
14.6.	Deployment device	26
14.7.	Back up	26
14.8.	Software testing	28
14.8.1.	Unit test	28
14.8.2.	Component and integration test.....	28
14.8.3.	Functional test	28
14.8.4.	GUI test	28
14.8.5.	Acceptance test.....	29
14.8.6.	Performance test	29
14.8.7.	Security test	29
14.8.8.	Non-regression test.....	29
14.9.	Development phases and acceptance of a product	29
14.9.1.	Initialization and design iterations	29
14.9.2.	Development iterations.....	30
14.9.3.	Acceptance iterations	30
14.9.4.	Software deployment and running	30
14.10.	Defect Handling.....	30
14.11.	Other quality commitments	31
14.11.1.	Design quality.....	31
14.11.2.	User load adequacy	31
14.11.3.	Non regression between versions	31
14.11.4.	Compatibility of the products with all system land scape platforms	31
14.11.5.	Documentation	31
14.11.6.	Code documentation.....	32

14.11.7.	Specifications consistency.....	32
14.11.8.	Test Datasets	32
14.11.9.	Traceability Matrix	32
15.	Software development plan.....	32
15.1.	Setting up the platforms	33
15.2.	Development teams	33
15.3.	Development Plan.....	34
16.	Infrastructure development plan	34
17.	Training plan	34
17.1.	End use training	35
17.2.	Administration Training (ADM).....	36
17.3.	Information security training (1ST).....	38
17.4.	Operational assistance training (OAT).....	38
17.5.	Training time schedule.....	39
18.	Project management.....	39
18.1.	Project organization	39
18.2.	Project communication plan	42
18.3.	Change control procedure.....	43
18.4.	Project quality assurance	44
19.	Risk management plan	45
20.	Project Product Description and Deliverables	48
21.	Project Phasing and Tentative Timeline	48
22.	Warranty services	50
23.	Post warranty services.....	53

List of Table

<i>Table 1 High level requirements.....</i>	<i>8</i>
<i>Table 2 Project team Role and Responsibilities.....</i>	<i>41</i>
<i>Table 3 Project issues classification type.....</i>	<i>43</i>
<i>Table 4 Project issue severity classification</i>	<i>44</i>
<i>Table 5 Risk Matrix</i>	<i>47</i>
<i>Table 6 Project planning and monitoring system deliverables</i>	<i>48</i>
<i>Table 7 Project Phasing and Tentative Timeline</i>	<i>48</i>
<i>Table 8 Issue Type</i>	<i>51</i>

<i>Table 9 Severity level of issue</i>	<i>52</i>
<i>Table 10 Handling of issues</i>	<i>52</i>
<i>Table 11 Issue response time and type</i>	<i>53</i>

List of figure

<i>Figure 1 High level system architecture</i>	<i>12</i>
<i>Figure 2 Development ,QA and Staging deployment diagram</i>	<i>27</i>
<i>Figure 3 Information security management process.....</i>	<i>Error! Bookmark not defined.</i>
<i>Figure 4 Project Organizational Structure</i>	<i>40</i>
<i>Figure 5 Risk management procedure</i>	<i>46</i>

1. Introduction

1.1. Background

There are multi-faceted problems related to land administration and management. Problem in provision of serviced land for investment and other uses, implementing urban plans, controlling illegal settlements and providing security for property holders are just few examples. But the prevalence of pervasive corruption is the most serious problem. Land especially urban land has been identified as one area where rent seeking activity is highly prevalent. One of the mechanisms to effectively combat rent seeking is establishing an effective land administration and management system.

The strategy adopted by the government of Ethiopia is follow a step by step or evolutionary approach towards creating an effective and efficient land administration and management system. According to this strategy first a cadaster and land registration system or legal cadaster has to be built then other systems such as property valuation and taxation will follow based up on the foundation laid by the legal cadaster. This approach is the best but which is not an efficient for us, still the problems are not resolved. The corruption, informal settlement, inefficient service delivery, tenure insecurity, land information data insecurity, disorganized land information, customer dissatisfactions etc. still our land administration service big problems and also the implementation cadaster going slowly.

Implementing the legal cadaster system and then upgrading the legal to multipurpose is an ideal solution and which is the only option to have a sustainable development and to implement a modern LIS but in our country current context focusing only on cadaster is not an efficient approach to transform our country land services to the proper level, there are so many bottlenecks around the cadaster implementation, the main problem is dis integrated land service and dis organized land information accusations, adjudication process in efficiency , data quality etc.

Each land stockholders' collaboration is not well and also we have not any enabling environments to practices the stockholders' collaborations. So we have to build an integrated land service, all land sector business process has to be automated and integrate each other.

We have a better experience to develop and practice the legal cadaster but there has to be a system that systematically facilitate the land administration services before the cadaster business process. If we practice an integrated land service, we can make the tenure secure, the land information accusation organized and secure, the adjudication process efficient and the cadaster implementation more efficient.

The main objective of the delivering integrated land service is assuring tenure security, delivering clear and efficient service delivery. So, apart from providing tenure security and facilitating the establishment of efficient property markets, integrated land service system will provide the base or fundamental data required by various stakeholders such as banks, courts, utilities, tax offices, other land related sectors to name a few. So, the integrated land service delivery system is expected to serve the above mentioned objective.

Oromia national regional state land bureau(ONRSLB), Oromia ICT Agency(OICTA) and Different ONRSLBs' cities has been working so many tasks to give a good service on the land sector, this indicates the dedication of the Oromia national regional state(ONRS) to becoming good governance in the sector. As a part of our country's government, INSA also has an interest in the land sector. The main goal of INSA in the sector is to establish a secure and reliable land information system platform, to have sustainable development for our country, and to keep our national security going in the right way. So we have the same interest as ONRS.

INSA has made some assessments of the current activities and we would have to appreciate /respect ONRS's current dedication and interests. But as per our assessment, the work is not organized, systematic, and integrated. The efforts and resources are not centralized because of this there is resource wastage and duplication of efforts.

INSA is a governmental institute that is responsible for securing the cyberspace and Digital Transformation of Ethiopia. Founded on proclamation NO.808/2013 of EFDR house of people's representatives, INSA has the exclusive mandate to standardize cyberspace standards and cryptography of the country. With more than 10 years of experience in developing enterprise systems and standing responsible for securing Ethiopia's cyber security, the institute has been providing many vibrant solutions for many business processes. The previously developed and implemented legal cadaster by INSA for the regional state under the ministry of urban and construction can be mentioned as an example and used as a basis for the Integrated land service delivery system/ILS/. As INSA is enrolled in automating many federal and regional office business processes, the institute has now planned to comprehensively study and develop an enterprise automated system for Oromia integrated land service system project.

1.2. Document purpose

The purpose of this document is to provide an establishment for the initiation of the ILS Project and to present a project definition and outline of the intended project. In addition, it is aimed to provide a way into the ILS project and acts as a first step to producing a Project Initiation Document. And also, it is a statement that describes the project cost, objective, scope, timeline, interfaces, and constraints of a project.

1.3. Audience

This document is intended to be read and used by

- Client (ONRSLB and ONRSIA)
- ONRSLB s' Cities
- ONRSLB s' Sub cities
- ONRSLB s' Keebles
- Project implementer (INSA)

1.4. Objectives

The general objective of this project is to design, develop and implement integrated land service delivery system for Oromia national regional states'.

The following are Specific Objectives of ILS:

- Document handling and property registration
- Online Service Delivery
- Integrate DERASH Payment Platform
- Implement a Subsequent Registration service delivery system
- Integrate ILS with existing Systems'
- Develop Data Migration tool
- Land Bank
- Empowering regional and pilot sites' ICT professionals on selected competencies

1.5. Project Scope

The project is scoped to automate Integrated land service delivery system for Oromia National Regional States' at regional, city and Keeble level.

1.5.1. In Scopes

- ILS Development and Maintenance
- **Requirements analysis:** includes defining and modeling a standardized business process for the region, city, sub-city and Keeble level land service offices, defining the requirements of the system and documenting the results in a document.
- **System architecture design:** includes identifying the different components of the system and the interaction that exists among them.
- **Developing ILS systems:** developing the systems that exist at region, city, sub-city and Keeble level components of the system namely
 - ✓ Document handling and property registration
 - ✓ Online Service Delivery
 - ✓ Integrate DERASH Payment Platform
 - ✓ Implement a Subsequent Registration service delivery system
 - ✓ Integrate ILS with existing Systems'
 - ✓ Develop Data Migration tool
 - ✓ Land Bank
 - ✓ Empowering regional and pilot sites' ICT professionals on selected competencies
- **Training:** Training of end user, system administrators and managers
- **Support:** Providing support to help desk personnel as well as the second level support that exists at the center, which is the ministry of urban development, housing and construction ministry.
- **Warranty and post warranty services:** Providing warranty services for a duration of 6 months as well as a post warranty service for a duration of 6 months.
- **Information security related services**
 - Designing information security management framework
 - Conducting information security awareness programs
 - Conducting security evaluation and auditing

- **Training**

- Prepare operational manuals and training manuals
- Train the end users, systems administrators and managers how to operate and administer the systems that are going to be deployed

1.5.2. Out Scopes

- Activities done while project initiation
- Goods/service purchase process
- Across project resource allocation
- Activities done while project closing
- Operating the system at the ILS

Any works that have not been stated in the scope are considered out of the scope of the project.

1.6. Project Assumptions

Assumptions that can be expected in the project development process:

- The needed resource will be available throughout the project time line
- There has to be a dedicated project follow-up from client-side
- There will be good communication between all stakeholders.
- ONRSLB will be a willingness to work with us, during all processes of the software development cycle.
- ONRSLB will assign personnel to work with us in each system deployment.
- We have an expectation of a stable business process and regulation used by ONRSLB.
- A change control procedure is required, so changes are only made when the cost is worthwhile.
- Incremental delivery of versions will provide the experience of using the system, which will help the customers to identify the real requirements.
- ONRSLB is responsible to arrange worded net network infrastructure and server infrastructure with a backup.

1.7. Project Constraints

The following are the possible constraints that are expected in the development of integrated land service delivery system (ILS).

- The main constraint is time; our client needs the product to own within a short period of time, on the other side we didn't compromise the software quality so we need time to develop the ILS with high quality. so if ONRSLB doesn't give us an appropriate time the maintenance cost and time will be increased.
- Employee turnover from the contractor and also from the sponsor side may affect our development time.
- In our country current context, higher officials changed from time to time. When the new officials come, they may initiate the work from start and it may also affect our delivery time.
- Lack of available WAN infrastructure and enough bandwidth required by the system
- Difficulty of involving city, sub-city and kebele representatives in the implementation of the project
- Lengthy procurement and vacancy process

1.8. Project tolerances

Constraints mentioned above can be tolerated if it does not affect the activity and budget schedule detailed in a formal plan and architectural design. If the business process reforms are leads to deviation from the activity and budget schedule detailed in a formal plan; then renewal of the contractual agreement is mandatory which requires additional budget and time to undertake the project.

2. Project Impact

The impact of the execution of the automation project is:

- Enhanced Cities' and Regions' efficiency by providing reliable land information.
- Increased Cities' and Regions' managerial efficiency due to informed decisions by providing dashboards and reports.
- Decreased decentralized land services tasks by providing a secure, reliable, and high-performance system.
- Increased land service delivery efficiency.
- Increased integration and communication among work units and regional offices.
- Enhanced communication and reporting among higher officials.
- Increased management decision-making capability through actual data.
- Increases investment attraction by making information availability and bureaucracy line simple.
- Increased customer satisfactions
- Increased cities' income
- It supports the intended political-economic development of the Oromia regional state.
- To have citizens' Tenure Secured
- Secured Land Information and Service delivery
- To enable Oromia regional land information integration with other sectors like financial institutions, the regulatory body, document authentication, and authorization, Land Planner, Ethiopian road authorities, and other stakeholders.

2. Overview of the project (ILS)

This system will maintain the need of integrated land service delivery system/ILS/ at regional, city and sub city and kebele level. ILS is not a single system and its composed of ILS Document handling system/IDMS/, ILS delivery system, Online service and Payment and ILS Data Migration/IDMT/.

There are there deployment options

- Having a centralized system at regional level providing services. Though this option requires a stable connectivity between regions cities, sub cities and kebels from operational and implementation perspective its very less costly and easy to manage.
- Having fully decentralize system. This means every city will have its own system deployed locally at their premises.
- A Partially decentralized system, with one centralized system located at region level and where there will be city level read only system used for doing some operations when the network is not up and running

Our option of the above mentioned options we proposed the centralized option to be adopted for the implementation of the ILS.

9.1. High level requirements

Table 1 High level requirements

Level	Descriptions	Requirements
Regional	• Will have an aggregated Land Information System. • Will have a web based service delivery system • Online payment and integration with ILS • Online service delivery • Will have web an interactive reporting • Will have a document management	1. Document handling and property registration ✓ ILS shall provide the mechanism to store the Ownership Source Documents and Applicant source documents digitally. Subject to revision the documents that need to be digitally stored are: • Ownership Source Documents • Applicant Source Document ✓ Subject to revision the document types

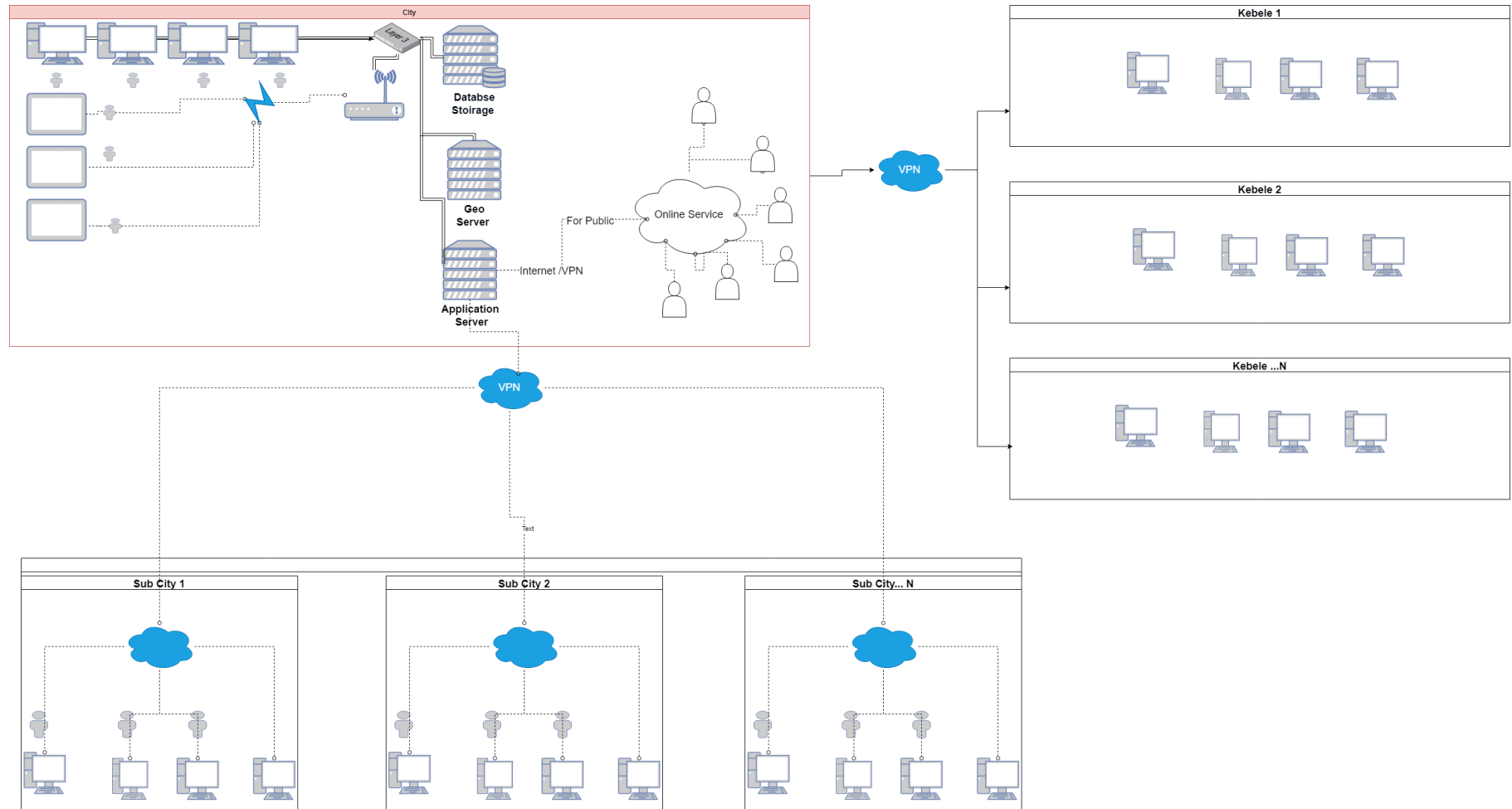
	system and integrated with ILS. • Will provide interface for integration with other systems	supported by ILS are: • Pdf files • Image files (such as .jpeg) ✓ Major Functionalities are • Upload document • Scanner integration • Document enhancement capability • Document preview • Search of documents • Integration with other components of ILS
Cities	• Will have a web based service delivery system • Online payment and integration with ILS • Online service delivery • Will have web an interactive reporting • Will have a document management system and integrated with ILS. • Will provide interface for integration with other systems	2. Online Service Delivery ✓ ILS has a mechanism to delivering a service online through an internet. The core business is not exposed on the cloud, but the customer can issue their application through the internet.
Sub-City	• Will have a web based service delivery system • Online payment and integration with ILS • Online service delivery • Will have web an interactive reporting • Will have a document management system and integrated with ILS. • Will provide interface for integration with other systems	✓ Major functionalities • Search properties through certificate number and view specific detail information • Issue an application for the selected service type • Upload necessary documents • View application detail • Cancel Application Detail • Print acknowledgement receipts

Kebele	• Will have a web based service delivery system • Online payment and integration with ILS • Online service delivery • Will have web an interactive reporting • Will have a document management system and integrated with ILS. • Will provide interface for integration with other systems	• Track application status 3. Integrate DERASH Payment Platform • Transfer payment detail to DERASH Payment platform • Pay the payment using the application registration number using the mobile APP and another mechanism. • Check whether the payment paid or not • Payments Dashboard 4. Implement a Subsequent Registration service delivery system The core business is giving an integrated land service delivery. After digitizing the document and registering the detail of the document (Party, Right and Restrictions), there will be a subsequent registration of land service delivery. ✓ Major land services are • Prepare new title certificate • Replace title certificate • Register mortgage • Cancelation of mortgage • Modification of mortgage • Register court injunction • Cancelation of court injunction • Modification of court injunction • Modification of ownership information • Split land
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		• Merge land • Land expansion request • Information extract • Transfer of ownership ✓ Major functionalities for all services are • Application registration • Acknowledgement and Rejection receipt • Document preview 5. Integrate ILS with existing Systems' • Integrate with existing DMS systems • Integrate with CRPRS 6. Develop Data Migration tool • Validate data • Migrate spatial data 7. Land Bank • Land request from customer • Review request • Land preparation • Decision making
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9.2. High level system architecture

Figure 1 High level system architecture



10. Project stakeholders

The development of this project involves the following key stakeholders

- Information network security administration (INSA)
- Oromia National regional state land office(ONRSLB)
- Oromia national regional state ICT Agency (ONRSICTA)

11. Project Approach

According to the need of ONRSLB, INSA will design and deploy integrated land service delivery system. Network infrastructure and servers that runs the developed application are provided by the customer (ONRSLB and ONRSICTA). The applications will be developed from the scratch and the following methodologies will be employed in application development to acquire the requirements of the systems.

Joint Application Development (JAD): Software developers, End users and the experts from the organization will work jointly to identify the functional requirements. This can be achieved through document analysis, direct observation, and interviews. While developing this system, the following activities will be performed

- Gathering requirements from responsible ONRSLB and ONRSICTA Professionals.
- Presentation will be prepared on each output for ONRSLB responsible bodies.
- Reports will be prepared on each phase by the project manager to ONRSLB responsible body.
- There will have knowledge sharing between ONRSLB and ONRSICTA assigned professionals and the system development team.
- We will apply agile software development methodology so every 15 days there will be a product demonstration,
- ONRSLB project office professionals can see the progress of the development every 15 days and can give feedback.
- Testing will be prepared before final deployment.
- Training will be prepared before final deployment.

12. Requirement Analysis

Requirement analysis is the starting point of any software development process. At this stage the requirements of the ILS at regional, city, sun-city and kebele will be identified and defined. The requirement analysis will cover the following major interrelated activities: -

- **Analyzing the existing system:** covers understanding the existing organizational structure, business process, legal framework (law, regulations, directives, and standards) and identification of risk associated with information security. The risks related to information security will be identified based on internationally known standards such as ISO and NIST.
- **Proposing the future system:** The proposed system will clearly show both the business and system requirement of the system. The requirement will include both functional and non-functional requirements. The proposed system will also include a standardized business process and organizational structure of the region and City ,Sub-City and Kebele of entire Oromia regional state. The business process and organizational structure that will be designed will contain control measures to ensure information security. In addition, defining the role and responsibility of users and administrators of the system will also be part of the proposed system.

The methodologies used during requirement analysis mainly constitute holding discussion, reviewing exiting documents and conducting field visits. The methodologies are described in the detail here below.

Discussions: After official commencement of the project a two or three days' workshop session will be conducted with stockholders. This will be a discussion that will be facilitated by a product owner assigned by INSA. In this discussion initial high level requirements and risks will be identified and prioritized. The result of the discussion will be submitted to the Oromia regional state land office as a form of initial version of the product backlog containing list of user stories along with their priority level and acceptance criteria. The costs for the workshop will be covered by the INSA.

Review of existing documents: The business understanding document produced by INSA and the laws, regulations, directives and standards prepared by the Oromia regional state regional land office will be reviewed in detail. From this possible system level requirement can be extracted.

Field visit: Since there are no operational land use offices in all administrative boundaries, with the exception of some pilot sites, it is assumed that field visits may not greatly help in identifying the system requirements. But nonetheless existing land offices of one selected as well as the regional land office and other regional stockholders will be visited to understand the operating environment of the system. This will -greatly help in identifying risks related to information security. Once an understanding is made about the existing system, INSA will analyzing define the requirements of the system. In addition, INSA will also clearly define the risks and required control measures.

Before the final requirement analysis document is submitted for approval to the client, the findings of the requirement analysis will be presented to the client.

The entire system requirement analysis will be documented in a document called the requirement analysis document (RAD). The contents of the document will be based the standard used by INSA for similar projects. The requirement analysis document will contain an overview of the existing system, the process and methodology conducted to review the existing system and the proposed new system, a description of the proposed engineered or reengineered business process and organizational structure, list of functional and non-functional requirements along with their unique identifier, description and quality criteria used for acceptance. The business process will be model using BPMN2.0 and enterprise architect software tool. In addition, to the printed document all the models prepared using enterprise architect will be delivered to the client.

In addition, detail risk assessment and control measure conducted during the requirements analysis will be delivered as separate document. Even if some of the security control measures are include in the requirement analysis document imbedded in the proposed organization structure, business processes and non-functional system requirements detail description of the risks along with their control measures such as procedures and information security awareness programs will be delivered as a separated document. The detail examination of the available network and data center infrastructure will also be delivered as a separate document at the end of the requirements analysis stage.

There is a possibility for refining the requirements after the requirements analysis document is approved by the ONRSLB. But completely new requirements should not occur; if new requirement(s) are raised INSA and the ONRSLB will discuss on how to handle such new requirements. One possible way for handling such change is through the issuance of change request

by the ONRSLB to INSA. The change request will be analyzed for the possible impact on the project cost, time and quality. INSA will use issue tracking system for handling issues that will arise during the project implementation. A tool named JIRA will be used for the requirements management aspect of the system.

13. System architecture design

Here we discuss mainly on the software architecture. Because the network and hardware infrastructure part of the system architecture is discussed in the infrastructure section of this proposal.

The software architecture for a system is the structure or structures of that system, which comprises software elements, the externally-visible properties of those elements, and the relationships among them. The definition of software architecture makes it clear that systems can and do comprise more than one structure and that no one structure holds the irrefutable claim to being the architecture.

Modern systems are more than complex enough to make it difficult to grasp them all at once. Instead, we restrict our attention at any one moment to one (or a small number) of the software system's structures. To communicate meaningfully about architecture, we must make clear which structure or structures we are discussing at the moment - which view we are taking of the architecture. Thus, the document of the software architecture follows the principle that documenting software architecture is a matter of documenting the relevant views and then documenting information that applies to more than one view.

Before commencing work on preparation of system architecture design a decision on whether to make the regional level centralized, decentralized or semi decentralized with read only access at the City ,Sub-City and Kebele level has to be made. The decision will be based on the findings of the analysis of the existing network infrastructure conducted during the requirements analysis stage. INSA will hold a two-day workshop with possibly with the domain experts, IT and GIS professionals on the different high level architectural options that can be used for the system. The decision for a specific architectural option will be based on specific principles. One among them is cost efficiency.

Cost efficiency can be seen from different perspectives.

These are

- **Purchase:** cost for the purchase of base software such as DBMS and operating system as well as cost of hardware.

- **Operation:** operational costs contain system administrators, the network cost, updates and so on. A well designed system can help reduce such costs especially the administration can be simplified by standardization and clear structures.
- **Maintenance:** since the system is expected to be operated in all urban areas for the coming years possibly decades the cost of maintenance can be reduced by following standards.

In addition, the most important factors to be considered during this stage are

- Efficiency
- Security (confidentiality, integrity and availability)
- Scalability: the system will start operation in six cities, but in due time it will be running in more than a dozens of City ,Sub-City and Kebele s.
- Maintainability: the architecture must be designed in a way that future requirements can be fulfilled by the system.
- Interoperability: web standards such as SOAP, XML, JASON, OGC specifications as well as other standards such LADM will be used.
- Standardization: in addition to the above mentioned standards standardization in naming conventions of hardware and software services, installation paths, log path, naming convention for users, required system services such as LDAP, and standard accounts for administrators also need major attention.

With regard to the implementation technology our choice is to implement the applications and database using open source technologies due to cost reasons, and also avoiding technological dependence on a specific vendor. The developer will choose the specific implementation technologies, and the client will be required to accept them. Once a decision is made on a specific architectural choice work on the system architecture design will proceed. As a final deliverable of the system architecture a software system architecture document for the entire ILS will be delivered to the client. The architecture and design document for the network and data center infrastructure will be delivered separately. The system architecture document will be based on ANSI/IEEE 1471-2000, also adopted as ISO/IEC 42010:2007, the recommended best practice for documenting--the architecture of software-intensive systems standard. Based on this standard the various concerns, of stakeholders will be documented using views. The following are the example of some of the views that could be included in the document

- **System context view:** All systems exist in some larger environment and a context view aims to answer questions about the environment and especially the technical relationships that the system being designed has with the various elements of the wider environment. The system is shown as a box and its interactions with external entities (other systems and actors). The viewpoint shows the external entities and their responsibilities that the system interacts with. A view conforming to this viewpoint describes a system from outside.
- **Deployment view:** Views conforming to the deployment viewpoint describe the mapping of software components to the hardware of the computing platform (physical distribution) and to third-party software server applications (e.g. web server). They show the environment into which the system will be deployed, including the dependencies that the system has on its hardware environment and details of the underlying network. In addition, this viewpoint shows where network connections are needed for the communication between the distributed software components.
- **Module decomposition view:** Views conforming to the module decomposition and use viewpoint partition the system into a unique non-overlapping set of hierarchically decomposable implementation units (modules) and shows how system responsibilities are partitioned across them. Hence the views show how modules depend on each other in the form of which module uses another module.
- **Data model view:** Data stored in a database are usually the biggest assets of a system. Views conforming to the data model viewpoint describe the static structure of the system's stored and transient data entities (e.g. databases and schemas) and their relationships.
- **Run-time view:** Views conforming to the runtime viewpoint depict elements that exist at the runtime of a system and the behavior and interaction of them. They clarify, which elements communicate with each other and the kind of interaction, for e.g. Synchronous or asynchronous. The elements could be associated with several computer processes or threads. The views illustrate how a system or part of a system works and thus how the different use cases are executed.
- **Security View:** This view illustrates how the entire software structure and associated security mechanisms fare against the threat model. It is the depiction of the solution as seen by taking security as the main point of focus. It shows possible points of attack, possible types of threats and their probability, weakness links and the overall robustness in the face of attacks.

UML diagrams will be used to represent the views diagrammatically. For example, UML deployment diagram will be used to represent the deployment view. Enterprise Architect software will be used for designing and maintaining the diagrams.

14. Software QA/QC/test plan

14.1. System landscape

14.1.1. General overview

The "system land scape" must be understood as the sum of all systems and applications' necessary for developing and exploiting future ILS. The system land scape contains the following platforms set up both in INSA and the client. Platforms that will be set up by INSA.

- **Development platform:** it is the core platform. The development platform mainly contains version control and continuous integration. GIT will be the software that will be used for version control; while Jenkins/team city will be used for continuous integration. The version control and continuous integration will have separate machines either physically or logically. The development platform also contains the development tools used by developer e.g. Visual Studio code community edition along with the necessary plugins. The development platform will not be accessed by the remotely over the internet by the client.
- **Test platform:** will be used for conducting both automated and manual system testing. The test platform will contain tools for automated system testing such as GUI testing. The client will be provided a remote access to the test platform. In addition to the testing platform that will be prepared by INSA during the software development, there will be a testing platform that will be set up by INSA at the ONRSLB to be used by the second level support personnel.
- **Production platform:** is the actual platform that will be set up by at the premises of the regional real property registration agencies. As part of the production platform here will be dedicated servers used for training purpose. These training servers will be used for training end users during the project period and for future training of users during the operational phase.
- **Software development management tools:** JIRA will be used for requirements management and bugs track.
- **Pre-production platform:** is a platform that is almost -identical to the production platform.

14.1.2. Sizing of each platform

The following specification is provided for each platform. Nonetheless, these specifications might be changed later on after an agreement is reached with the client.

Development platform

- **Database:** Linux Ubuntu Server quad core 16GB Memory, 2TB disk size.
- **Application Server:** Linux Ubuntu Server quad core 16GB Memory, 100GB disk size.
- **Geo Server:** Linux Ubuntu Server quad core 16GB Memory, 100GB disk size.
- **JIRA Server:** Linux Ubuntu Server quad core 16GB Memory, 100GB disk size.
- **Test link:** Linux Ubuntu Server quad core 16GB Memory, 100GB disk size.
- **Alfresco Server:** Linux Ubuntu Server quad core 16GB Memory, 2TB disk size.
- **Version control:** Linux Ubuntu Server quad core 8GB Memory, 100GB disk size.
- **Continuous integration:** Linux Ubuntu Server quad core 8GB Memory, 100GB disk size.
- **Test platform:** similar to the continuous integration server.

14.2. Source code quality

14.2.1. Code documentation

The main programming language that will be used by INSA is C# and the code will adhere to C# Coding Standard of INSA. INSA developers will write comments at the beginning of every complex algorithm, every methods and every class. At the end INSA will provide full code documentation.

In each phase the QA team will assure the coding standard is followed and a proper commenting is applied before each algorithm, method and class. In addition to this, QA will verify full C # doc is written for each system.

14.2.2. Programming style

INSA will follow the C# and JavaScript Coding Standard for modules/components: of system that will've developed using C#. For other programming languages industry best practices will be followed. The coding standards define

- The maximum number of function parameters, line lengths;
- Naming conventions for variable, object, class, interface ...
- Presence of mandatory headers

- The use of packets imports, of classes, of scope modifiers and of instructions blocks
- The spaces between some characters
- The good practices of class construction
- Code complexity
- Duplication of code
- Vulnerability

14.2.3. Code quality

Different tools or plugins, such as ESLint, sonarqube, and resharper, will be used to regularly analyze the source code in order to check for

- Programming style
- Unit test source code coverage
- Documentation code coverage
- Common bug-prone coding bad practices
- Architecture bad practices
- Code complexity

14.3. Software architecture

The following processes are defined for the architecture design phase.

- Define architectural principles
- Propose high level architectural options
- Conduct discussion and choose the most appropriate architectural option
- Perform architectural design and documentation

The following architectural principles are set.

- Efficiency
- Security (confidentiality, integrity and availability)
- Scalability: the system will start operation in 19 cities, but in due time it will be running in more than a dozens of City, Sub-City and Kebeles.
- Maintainability: the architecture must be designed in a way that future requirements can be fulfilled by the system.
- Interoperability: web standards such as RESTFUL, SOAP, XML, JSON, OGC specifications will be used.

- **Standardization:** in addition to the above mentioned standards standardization in naming conventions of hardware and software services, installation paths, log path, naming' convention for users and standard accounts for administrators also need major attention.

The QA team will verify if the processes are correctly followed and the documentation is based on the IEEE standard. Architectural review processes conducted by the QA team are more or less based on a discussion held with the software development team and review of the products.

14.4. Development platform

14.4.1. General overview

The development platform will be used for all current and future development activities of NCRPRS. Future development activities include releasing of new versions of the system, releasing of patches for errors encountered during the warranty service time and other similar activities. The development platform mainly contains a version control as well as continuous integration and automatic deployment servers and tools. In addition, the IDE that will be used by the developers can also be considered as the development platform.

The development platform will only be accessible internally by the system developer (INSA). The client will not be provided remote access to the development platform. The system developer (INSA) will also implement a disaster recovery and backup strategy. Backup strategy for all repositories, and disaster recovery plan in case the server failed, or in the time the server is not accessible to developers.

14.4.2. Development platform software

The development platform will include the following software tools installed

- **The GIT Lab server** for source code repository and version management
- **Jenkins** for Continuous Integration
- **Active directory** will be used for user management
- **ERGA (INSA mail server)** will be used for communication
- **Development tools** Visual studio code and other IDEs that might be required along with the necessary plugins for unit testing, coding standard checking and other similar purposes.
- **Unix** will be the operating system

- **Docker engine** for containerization of development tools and deployment
- **Test link** for test case management
- **Geo server** map publisher
- **IIS Server application** server

14.4.3. User types and roles

The main user types and roles are

- Git server administrator/gate keeper - responsible for setting up and administration of the version control
- Build master - responsible for continuous integration, deployment and release management. The build master will write different scripts to automate most of the work.
- Developer - responsible for
 - Check in and check out source code
 - Manage configuration files
 - Write and execute unit tests

Modification to the user types and roles can be made during the project execution period when necessary or when the need arises.

14.4.4. User registration

Users on the project will directly be registered on the Microsoft active directory database. After user account is created then the user will be prompted to change his/her password. Using a combination of LDAP, and Windows Authentication, a single account will be used to access every development environment tools and resources. In case a password has been forgotten, a secure mechanism will be implemented in order for a user to obtain a new password.

14.4.5. Source code hosting

The source code hosting will have the following functionalities:

- Source code browsing and downloading
- Writing/Modification of the source code
- Conflict management
- Version management
- Web access to source code

14.4.6. Continuous integration system

The continuous integration system will have the following functionalities:

- Automated tasks of completion test, reporting and documentation creation.
- Automated testes
- Alerting on build results
- Build administration and configuration
- No limit of build for a source code management instance
- Possibility to choose between several predefined configurations
- Web interface for build browsing and administration
- Automatic deployment on test instances

14.4.7. Continuous code inspection system

The continuous code inspection will have the following functionalities:

- Multiple source code inspection criteria
- Complete history of indicator result to assess trends on code quality
- Pre- commit check
- Inspection automation

14.5. Test platform

The test platform will be used by internal QA/QC/test team as well as the client to test the different sprint/iterations as well as releases of the system. The test platform will mainly be used to perform system testing.

Test platform will have the following functionalities

- Test plan management
- Requirement management
- Test case management
- Manage user roles; like Guest, Test Designer, Senior tester, Tester, Leader, Administrator
- Generate different report
- Allow access to clients
- Bug tracking system integration

14.5.1. Test platform software

- TestLink test management tool
- Apache Jemeter performance testing tool
- Selenium functional testing tool
- JIRA for bug /issue tracking and agile project management

14.5.2. Performance testing

Performance testing tool will have the following functionalities:

- Load testing - It helps to identify the maximum operating capacity
- Stress testing - in order to test the system's response at unusually high or peak loads
- Volume testing -tests the system ability in an operating environment to process a very large amount of data

14.5.3. Functional testing

Functional testing tool will have the following functionalities:

- Identify functionalities, determine output based on the function's specifications, and compare of actual and expected outputs.
- Check whether the application works as per the customer need.
- Regression testing- new software bugs, or regressions, in existing functional and non-functional areas of a system after changes such as enhancements, patches or configuration changes, have been made to it.
- Smoke testing- is preliminary testing to reveal simple failures server enough to reject a prospective software release.

14.5.4. Bug tracker

Advanced bug tracker JIRA will be used to manage tasks inside development projects. Bug tracking system will have the following functionalities.

- Manages and maintains lists of issues
- Incident ticket system
- Distribution and assignment of issues to persons in charge
- Monitoring of handling, time spent and quality of work
- Ensuring the observation of internal processes by forced control with help of workflows

- Statistical analysis of the number of tickets
- Systematic collection of questions and answers for FAQs
- Assignment of a priority to each issue based on the overall importance of that issue
- Containing a detailed descriptions of the problem being experienced, attempted solutions or workarounds, and other relevant information
- Maintaining of a history of each change.

14.6. Deployment device

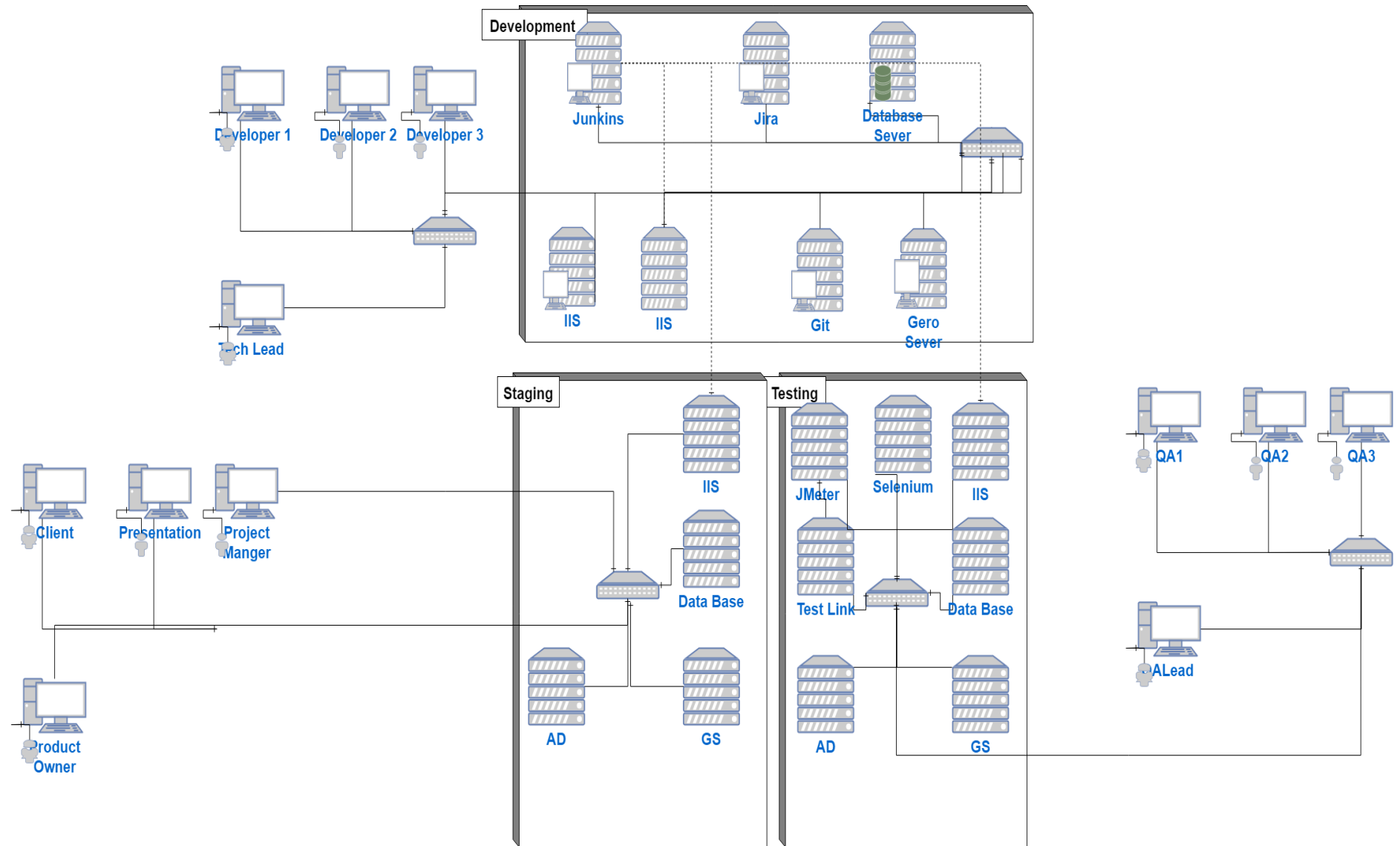
Any deployed version of a component is archived and some of the archived version is deployable only on windows environment while others can be deployable on any environment. The deployment device enables to:

- Register a deployment request
- Deploy any application resulting from a continuous integration build on any platform of the system landscape.
- Browse deployment history and status
- Access deployed applications log files
- Provide an integrated solution to deliver applications on any platform of the system landscape.
- Generate deployment report

14.7. Back up

Back up of all development platform and deployment device managed data will be taken once in a day. The following figure shows high level deployment diagram for automated testing framework.

Figure 2 Development ,QA and Staging deployment diagram



14.8. Software testing

Throughout the development process of the ILS system different types of testing will be carried out to ensure the quality of the system. Once the detail user requirement is delivered, the test plan with the detail test cases and approaches will be produced. But here we will define some of the general principles of the different type of tests.

14.8.1. Unit test

The programmers will write unit test. The tool that will be used for .NET Core developers is Nunit. The overall unit test coverage should be at least 80% and this is verified using a unit test coverage tool which will be decided later. The test coverage tool chosen will generate a test report with nthe test definition and test result for each build.

14.8.2. Component and integration test

Integration testing is when you test more than one component and how they function together. For instance, how another system interacts with your system or the database interacts with your data abstraction layer.

Component testing is fulfilled by programmers on the code written by them and with the support of development environment, such as a unit test structure or debugging tool Component and integration testing will be done by the system developer (INSA). Tests definitions and results will be available for TS and client review.

14.8.3. Functional test

The functionality of the system is tested at system level whether it performs and functions correctly according to design specifications. The automation of this test will be carried out using Selenium testing tool and test report with their results will be produced after each test.

14.8.4. GUI test

The QA team along with the client will check whether the interfaces designed and produced by the SD are as the specifications defined by the QA team and the clients manually. On areas where automation testing is required, Selenium testing tool will be used.

14.8.5. Acceptance test

This test is carried out at every software release after the defects are identified through the functional and GUI testing is corrected and the end user documentation for that specific release is produced. The client with the help of the QA team will conduct the acceptance test manually. The test data for this test can be provided and approved by client.

14.8.6. Performance test

A test plan with the detail test cases will be prepared based on the client requirement for the performance of the system. After the test plan is reviewed by the client, a performance test which can be repeatable will be carried out on the QUALIFICATION platform by the QA team using Jmeter testing tool. At the end of each performance test, test report with the test result will be delivered. Performance test will also be carried out on pilot project.

14.8.7. Security test

Different security tests will be conducted internally as well security tests will be conducted on the infrastructure and the final deployed system at the user premises. Maintainability, Interoperability, Compatibility and Reliability testing will be carried out by the developer and the report will be delivered to the client at the release of every software product.

14.8.8. Non-regression test

For each software product a non-regression test suit will be prepared and this test suit will be executed before deployment.

14.9. Development phases and acceptance of a product

Generally, the product will have four phases. Each phase is composed of iterations. The iterations will have the same duration.

14.9.1. Initialization and design iterations

At this iteration, the requirement and design of the integrated land use planning and monitoring system will be clarified and detailed in a document through discussion between the client and the system developer (INSA). And this document will be a binding document for the acceptance or rejection of the final software product.

14.9.2. Development iterations

Following the development plan, INSA will develop ILS releases iteratively. During iteration the developers will implement the functionalities described in iteration/sprint backlog. Developers will follow the coding standard during coding and they will write unit test simultaneously. At the end of iteration, the system will be ready for test by the QA team. The test result will be provided to the client either using the collaborative environment or officially through a letter. The bugs identified during testing will be rectified in the next iteration. And every release, following the minimum viable product approach, will be deployed on the client site for test. When the product is at the acceptable level, the product goes to the acceptance iteration. If not, the developer fix the bugs found and the above actions will be repeated until the product is at the acceptable level.

14.9.3. Acceptance iterations

At this iteration, the client will provide test plan and will check test report by conducting all the tests on. If the test reports of the system developer differ from the test reports of the client, the issue will be reported through the web portal and the system developer will fix it or resolve the issue. Test report will be approved by client.

14.9.4. Software deployment and running

After the client accepts the product, the product will be deployed on the production platform of the client by the system developer (INSA) and every bug that are reported will be reported through the issue tracking tool and the system developer (INSA) will fix the bugs accordingly.

14.10. Defect Handling

The following defect handling process will be followed.

- An issue is raised by the QA/QC/test team or the client
- The issue will be classified based on the classification criteria (type, severity and priority) described below. The classification will be done by all concerned parties
- The issue will be registered in the issue register using the defect tracking tool
- The issue will be handled according to its classification
- The issue register will be updated

14.11. Other quality commitments

14.11.1. Design quality

The system developer (INSA) is the sole responsible for:

- Functional and technical consistency of all the software components of the system
- Identifying the impacts of a required evolution of the system

The system developer (INSA) is then responsible for:

- Making sure that general and detailed specification he delivers for each component match with the business requirements of the client
- Detect the possible incompatibilities or impacts that could hinder the capacity of the solution to satisfy the business needs of the client
- Ensuring the functional and technical consistency between the specifications he writes, the requirements expressed in the requirements analysis document and other systems that might interact with the ILS system

Later changes resulting from system developer (INSA) negligence will never be considered as evolutions of the system and will be held as defect and corrected by the system developer (INSA).

14.11.2. User load adequacy

INSA will vouch for the fact the application he developed and deployed will be able to sustain the user load.

14.11.3. Non regression between versions

The system developer (INSA) will guarantee non-regression (functional, technical, -performance, usability ...) with regard to previous versions when deploying a new version of an application.

14.11.4. Compatibility of the products with all system land scape platforms

The system developer (INSA) will be the sole responsible for the compatibility of the developments with different platform of the system landscape. Every product must produce the same results on all the platforms.

14.11.5. Documentation

The system developer (INSA) will produce and maintain an exhaustive, clear and up to date document repository. The system developer (INSA) will be responsible for any necessary upgrade tasks.

14.11.6. Code documentation

The system developer (INSA) will produce a perfectly and completely documented source code. The SD will be responsible for any necessary upgrade tasks.

14.11.7. Specifications consistency

The system developer (INSA) is responsible and vouches for the consistency between the developments and the detailed specification that are delivered with it. In case of discrepancy the system developer (INSA) will be in charge of upgrade one or the other with regards to the client needs. The system developer (INSA) vouches for:

- The consistency between every detailed specification for a given version of the system
- The fact that detailed specifications cover every requirement provided by the TS and the client

14.11.8. Test Datasets

Creation and verification task of test datasets such as:

- Sampling of raw data
- Format conversion
- Data generation
- Consistency control of test datasets for development, test
- Update of test datasets will be the responsibility of the SD
- The client will be able to provide sample raw datasets.

14.11.9. Traceability Matrix

Traceability matrix is a document, usually in the form of a table that correlates system use cases and test cases to determine the completeness of the relationship. Microsoft Excel tool will be used for traceability matrix.

15. Software development plan

Once an approval for the software architecture design is acquired INSA will immediately commence with first developing the region level system followed by the City, Sub-City and Kebele level systems. The following major activities will be carried out in the system development stage.

15.1. Setting up the platforms

INSA will avail different platforms. These are development, testing, capacitation/training and pre-production platforms. The production platform is expected to be set up in cities. Setting up the production platform is described in the network, hardware and data center infrastructure section of this proposal.

- **Development platform:** it mainly consists of a version control installed and configured in a centralized server and client development tools, such as Microsoft Visual studio, to be used by developers. In addition, it will also contain continuous integration and automatic deployment servers. The client will not be provided an access to the development platform. See the QA/QC section of this proposal for detail configuration of the development platform, branching strategy.
- **Test platform:** various iterations and releases of the system will be deployed to the test platform to conduct system level testing. The platform will contain the deployed systems, tools for testing, test management and issue tracking.
- **Production platform:** is a platform that will be set up by the system developer (INSA) at the premises of the client.
- **Training platform:** is a platform that will be used for training purpose. The production environment will be used for training purpose.

15.2. Development teams

There will be one software development team responsible for the entire software development activity from requirements analysis up to development. This team has different sub teams. These are

- Requirements analysis team
- Software architecture design team
- ILUP System development team
- LIS development team
- Document management system development team
- Geo Portal Development team
- Quality Assurance team
- Development Operations(DevOps) team

15.3. Development Plan

Though the final and more detailed software development plan will be delivered following the acceptance of the software architecture documentation.

16. Infrastructure development plan

The server and network infrastructure is not the part of this proposal.

17. Training plan

Though a detail training plan will be produced and delivered to the client during the system development stage a brief description of the training is included in this section. These are end users (LIS and Document management systems), administrators (system and database administrators), first level and second level support personnel.

A detail training documentation will be provided to the client for approval one month prior to the conduct of training. For each course the training documentation that will be delivered to the client covers the following issues

- Objective
- Training Content
- Duration
- Number and Qualification of Attendees
- Location of The Training
- Resources Required for The Training

The ONRSLB is expected to provide the necessary resources, such as training rooms, digital projector and white board, required for carrying out the training. A detail resource requirement will be provided in the training plan document. The final deployed system (production environment) will be used for training. After the training the database will be cleared and made ready for operation.

INSA will maintain a register of all the trained personnel and deliver it to the client after the completion of each training course. INSA will use Ms Excel to maintain the list of trained personnel. The training report that will be delivered to the client will include the list of trained personnel as well as result of the assessment of the trainees. INSA will provide a certificate to the trainees. Based on the type of training

the training is categorized into four types. These are end use training, Administration training, information security training and operational support training.

17.1. End use training

The objectives of this training category are

- To enable end users to have the required knowledge and skills to work with the systems according to their assigned task.
- To enable the managers to understand the impact of the new system on their organization.

The training target groups are end users (operators of the system), Administrators, operational support personnel and managers of the real property registration offices at City, Sub-City and Kebele and region level. This training category is divided in two main sub categories. These are City, Sub-City and Kebele level training and regional level training.

17.1.1. City, Sub-City and Kebele level training

In order to meet the above mentioned objectives the following training courses are proposed for City, Sub-City and Kebele level training. The end user training conducted for the pilot site(s) will be an informal training. All end user trainings will be based on user manuals prepared by the system developer (INSA).

- End user training (EUT): all EUTs will be conducted at their respective site
 - EUT 1: Introduction of the legal background of the system
 - It is one-day training. Users will learn the more general concept of integrated land use planning and monitoring system and legal basis of the system. This training course is intended to all users of the region and City, Sub-City and Kebele level system.
 - EUT 2: General introduction to the system
 - It is a three-day training. Users will learn the basics of how to use the system. An overview of the entire system will be provided. The participants are expected to have a good knowledge of computer usage. This training course is intended to all users.

- EUT 3: ILS application usage
- It is a four-day training. Users will learn in depth on the features of the land use system. They will learn how the integrated land use application is integrated in to the entire business process.
- EUT 3: Document handling application usage
- It's a half day training. Users will learn how to digitize documents and integrate with other part of the ILS
- Management training (MT): Managers of the city level registry offices will share EUT1 and EUT2. In addition to these training courses the managers will learn
 - MT1: Understanding the tasks of IT administration
 - This will be two-day training. Managers will understand the task of IT system administration unit, the different professions need in the unit and the cost of maintaining the system. In addition, they will understand the impact of the system in the operation.

17.1.2. Region Level Training

The regional level end users will be share all of the City, Sub-City and Kebele level training.

- End user training (EUT): all EUTs will be conducted at their respective site
 - EUT 1: GIS
 - It is a three-day training. It will cover basic GIS as well as advance GIS concepts. This training will enable the end users to do different GIS analysis by integrating the cadastral map and other spatial information such as land use.
 - EUT 2: System usage
 - It is a three-day training. Users will learn on the features of the regional level system. At the end of the training users will be able to migrate data, modify land use data and use other features of the system.

17.2. Administration Training (ADM)

The target group of this administration training category are network and system administrators, and database administrators. This training category will be conducted centrally at a premises prepared by the client. As part of the operational support assistance that will be provided by INSA for a period of not less than four weeks, reinforcement training and close support will be provided.

The main objectives of this training category are

- To enable administrators to understand the general tasks of IT system and infrastructure administration
- To enable database administrators to carry out their day-to-day task
- To enable system and network administrators to carry out their day-to-day task

Even for administrators it is very useful to know the overview of the system and the legal basis. So the administrators will share City, Sub-City and Kebele level all end user trainings. The following courses will be provided in addition to the end user trainings.

- ADM 1: Basic administration
 - The system administrators will learn the components of the system and how they work together. It takes two working days. The administrators should have basic knowledge of networking and installation and configuration of server software.
- ADM 2: managing the application servers
 - In this course topics related to GIS server administration and web server administration will be covered. The training will take three days.
- ADM 3: administering the databases
 - The course will provide database administrators with the capability to administer the databases. This course will be specific to the technology used to implement the database. This is not a full administration course on the specific DBMS used. So the trainees should have a basic understanding and preferably a working experience of database administration (e.g. with PostgreSQL databases). This course will take four days. This training will cover issues related to the data model, the different database instances, replication, backup and recovery and other related items.

ADM 4: Server operating system, this training will focus on how the network and systems administrators will install, configure and operate the operating systems of the servers and storages, and backup systems.

17.3. Information security training (1ST)

Based on the designed information security management program trainings will be designed and conducted. With this regard the training will try to achieve the following objectives.

- Creating information security awareness in management and users
- Build the necessary technical capability of the organizational structure created as part of the information security management program

The detail content of the training program will be identified in the first phase of the project implementation period.

17.4. Operational assistance training (OAT)

The main objectives of this training category are

- To enable the help desk (first level support) personnel of the city level registry office understand their task and be capable in carrying out their responsibility
- To enable the second level support personnel located at the federal urban real property registration and information agency to understand their task and be capable in carrying out their responsibilities

Operational assistance training will be conducted centrally in one location to be arranged by the client. In addition, the operational support assistance services that will be provided by INSA for a period of not less than four weeks will be used to help the first level support personnel in creating the necessary operational capabilities.

City, Sub-City and Kebele level help desk personnel will share City, Sub-City and Kebele level EUT 1 and 2. In addition the following training course will be delivered to the help desk and the second level support personnel.

- OAT 1: understanding the support structure and procedure
 - This is two days training. Both help desk and second level support personnel will take this training. It consists of teaching the trainees on the support structure and procedure

described in the operation assistance section as well as the use of tools of reporting and monitoring on issues.

- OAT 2: typical system problems
 - This is three days training. It consists of teaching the help desk personnel on the major components of the system and the frequent problems that might occur as well as their resolution mechanism. The overall architecture of the system will be covered in this training course.

17.5. Training time schedule

The training schedule will prepare before the training session start.

18. Project management

We have changed the section from human resources plan to project management because this section covers other issues apart from human resources. In this section the different project teams along with their responsibilities, description of key personnel, project communication plan, change control procedures and project quality assurance issues are addressed.

18.1. Project organization

The project has a project management team composed of the project board and project manager, six teams responsible for the project deliverables, one project support team and one project assurance team. The project management team includes the project board, the project manager, project assurance and support teams. The project management team is responsible for the overall monitoring, control and directing of the project. The other teams are called the delivery teams, which are responsible for producing the project deliverables.

The table figure and table below described show the project organizational structure and responsibility matrix of each team.

Figure 3 Project Organizational Structure

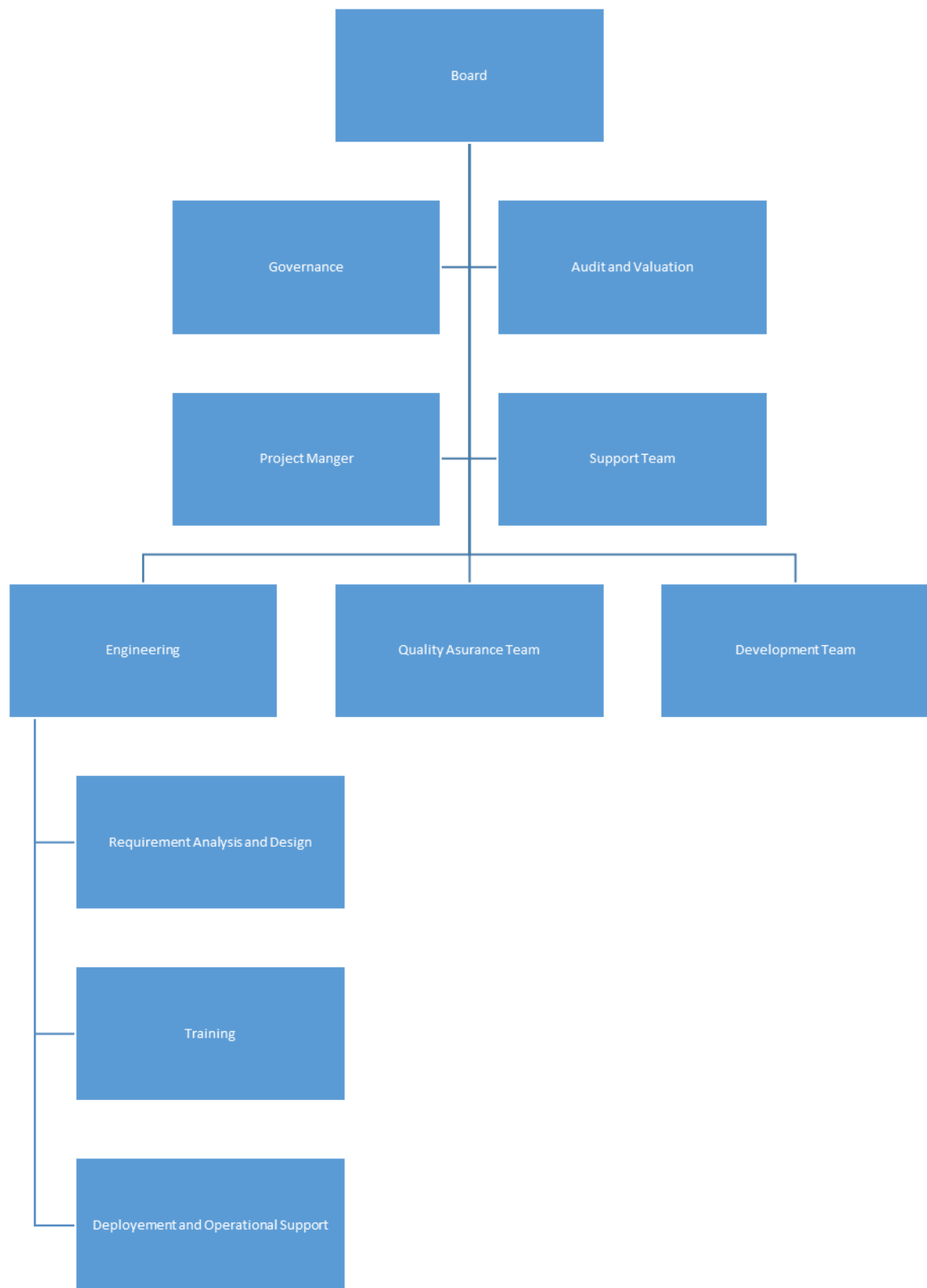


Table 2 Project team Role and Responsibilities

<i>No</i>	<i>Role</i>	<i>Description</i>	<i>Name</i>
1	Board	• <i>Responsible for the success of the project</i> • <i>Provide direction to the project (which is resolve issues and risks escalated, monitoring and control, approval of plans)</i> • <i>Take overall responsibility for communication issues.</i> • <i>Making sure the project processes and procedures are correctly followed and applied in the project.</i> • <i>Provide guidance on the application of the project processes and procedures</i>	<i>Miss. Tigist Hamid (INSA Deputy Director General)</i>
2	Project manager	• <i>Manages the day-to-day activity of the project.</i> • <i>Produce and keep up-to-date the project plan(s).</i> • <i>Submit project deliverables to customers</i> • <i>Produce reports</i> • <i>Serve as a communication point for the project</i>	<i>Tilahun Dessalegn (Cadaster System Division Head)</i>
4	Project support	• <i>Provide support in contractual issues to the project manager</i> • <i>Provide support to project manager in project documentation and other administrative issues</i>	<i>Dawit Abel</i>
5	Engineering	• <i>Responsible for producing the requirements analysis and design documents of the software components.</i> • <i>Prepare end user documentation and training manuals carry out training and evaluation activities</i> • <i>Provide onsite support to end users</i>	<i>Zelalem Teferi</i>

6	<i>Software Development</i>	• <i>Responsible for the development of the software components</i> • <i>Contains sub-teams for each component of the project (LUPMS, LIS, Geo-Portal, DMS and BI)</i>	<i>Mesfin Tafese</i>
7	<i>QA/QC/test team</i>	• <i>Set software quality assurance and control procedures</i> • <i>Conduct system testing on the software components</i> • <i>Produce test result reports</i>	<i>Tigistu Bekele</i>

18.2. Project communication plan

The project communication plan covers the type of communication, frequency of communication and mode of communication between the client and INSA.

- **Workshops**
- **Requirements analysis:** during the requirements analysis stage of the project a workshop will be held to identify preliminary requirements and another one will be conducted prior to delivery of final RAD.
- **System design:** a workshop will be held to discuss the different options of the software architecture.
- **Pilot system testing:** a workshop on the components of the system delivered for the pilot sites and how to use these components will be carried out.
- **Infrastructure deployment:** after the infrastructure deployment is carried out a workshop will be held to discuss the different activities carried out.
- **Final deployment of the system:** a workshop will be held with after the final system deployment.
- **Reports**
 - **Monthly progress report:** The exact content of progress report will be included in the inception report. The monthly progress report will contain information related to the project plan, pending issues, status of deliverables and other important aspects of the project.
 - **Mid project report:** a preliminary content of the mid project report will be included in the inception report and the exact form of the report will be submitted to the client two months prior to the report delivery date. Among other issues the mid project report will provide information on all consolidated results, assets transferred to the client and so on.

- **End project report:** The end project report will be submitted two months prior to the delivery date. The end project report will cover different issues. For example, detail warranty service plan as well as the post warranty service plan and financial proposal will also be delivered as part of the end project report.
- **Sprint planning and review meetings:** discussions will be held that will stay at most one full day at the end of every sprint with the client. The meeting will be conducted at the premises of INSA. On these meetings the previous sprint will be reviewed and a plan for next sprint will be discussed.

18.3. Change control procedure

Change is inevitable during the life of a project, and every project needs a systematic approach to the identification, assessment and control of issues that may result in change. Issue and change control is a continual activity, performed throughout the life of the project. Without an on- going and effective issue and change control procedure, a project will either become totally unresponsive to its stakeholders or quickly drift out of control.

The aim of issue and change control procedures is not to prevent changes; it is to ensure that every change is agreed by the relevant authority before it takes place. Change can only be considered in relation to an established status quo, i.e. a baseline. Therefore, a prerequisite of effective issue and change control is the establishment of an appropriate configuration management system which records baselines for the project's products and ensures that the correct versions are delivered to the customer.

Any issue that arises during the project lifetime will be classified based on its type and severity.

Issue Type Classification

Table 3 Project issues classification type

Type of issues	Definitions
Request for change	A proposal for a change to a baseline.
Off specification	Something that should be provided by the project, but currently is not (or is forecast not to be) provided. This might be a missing product or a product not meeting its specification
Problem or concern	Any other issue that the Project Manager needs to resolve or escalate.

Issue Severity Classification

Table 4 Project issue severity classification

Type of issues	Definitions
Critical	The change is essential for the viability of the project and must be addressed
Major	The change is useful but its absence does weaken the viability of the project
Minor	The change is neither essential nor important and can wait

All issues that will arise during the project execution period will be registered in an issue register. The purpose of the Issue Register is to capture and maintain information on all of the issues that are being managed formally. The Issue Register will be monitored by the Project Manager on a regular basis. The project support team will assist the project manager in maintaining and keeping the issue register up-to-date. The detail contents of the issue register can be provided in the inception report if there is a need for it. The issue register will be used during the project reporting period to report on issues. There will not be a separate issue report but a section of the period report will deal with issues that are captured and handled during the reporting period as well as the status of other issuer. INSA will follow the following issue and change control procedure during the project lifetime

- **Capture:** Receive issues, determine the type and severity of issues and register in-the issue register.
- **Examine:** Assess impacts on project objectives and check severity classification
- **Propose:** Identify, evaluate and recommend options.
- **Decide:** Escalate if beyond delegated authority, if not approve, reject or defer recommended options.
- **Implement:** Take corrective action, update records and plans.

The project board is responsible for taking decision on issues that have significant impact on the project duration, cost and scope. Other issues will be handled by the project manager.

18.4. Project quality assurance

INSA will use the internationally well know project management standard called PRINCE2. The processes and procedures described in the standard will be tailored to the current project context. The

project manager and others are expected to adhere to the project management processes and procedures such as risk management and change control. But it is the responsibility of the project assurance team to make sure the project is following the prescriptions and is being conducted according to the what has been set.

19. Risk management plan

Risk taking in projects is inevitable since projects are enablers of change and change introduces uncertainty, hence risk. Management of risk should be systematic and not based on chance. It is about the proactive identification, assessment and control of risks that might affect the delivery of the project's objectives. The aim of risk management is to support better decision making through a good understanding of risks - their causes, likelihood, impact, timing, and the choice of responses to them.

Management of risk is a continual activity, performed throughout the life of the project. Without an ongoing and effective risk management procedure it is not possible to give confidence that the project is able to meet its objectives.

A risk is an uncertain event or set of events that, should it occur, will have an effect on the achievement of objectives. It consists of a combination of the probability of a perceived threat or opportunity occurring, and the magnitude of its impact on objectives. There are two types of risks these are

- **Threat:** is an uncertain event that could have a negative impact on objectives
- **Opportunity:** is an uncertain event that could have a favorable impact on objectives

For risk management to be effective, risks need to be

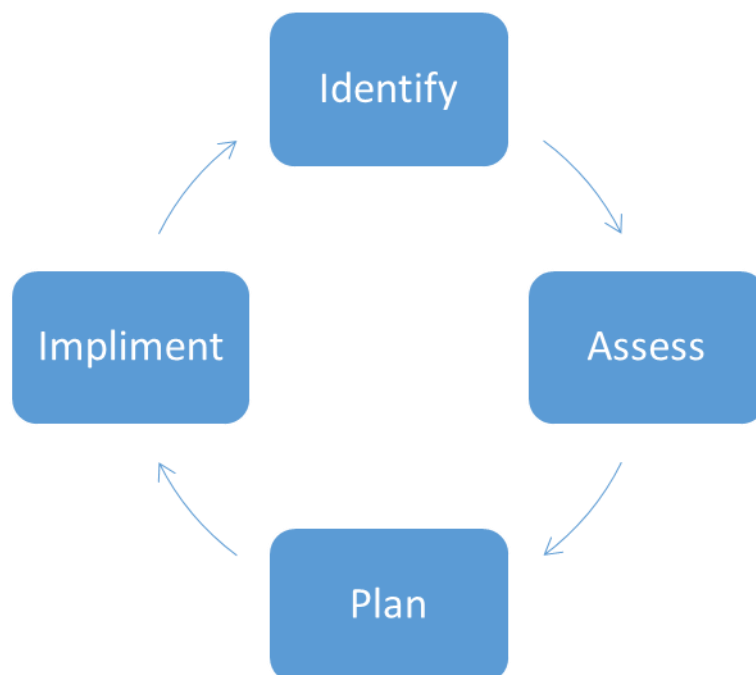
- **Identified:** This includes risks being considered that could affect the achievement of the project's objectives, and then described to ensure that there is a common understanding of these risks
- **Assessed:** This includes ensuring that each risk can be ranked in terms of estimated likelihood, impact and immediacy, and understanding the overall level of risk associated with the project
- **Controlled:** This includes identifying appropriate responses to risks, assigning risk owners, and then executing, monitoring and controlling these responses.

Since INSA uses PRINCE2 project management standard, the risk management procedure is also based on this standard. The risk management procedure comprises the following steps.

- **Identify:** identify the threats and opportunities that may affect the project objectives and register them in the risk register. One way of identifying risks is to conduct risk workshop.
- **Assess:** estimate the probability and impact of the threats and opportunities. The risk proximity will also be of interest to gauge how quickly the risk is likely to materialize if no action was taken. Evaluate the net effect of risks
- **Plan:** prepare specific management responses to the threats and opportunities identified, ideally to remove or reduce threats and to maximize opportunities. Possible threat responses types are avoiding, reduce, share and accept. Possible opportunity response types are exploit, enhance and reject.
- **Implement:** implement the planned risk response, monitor the effectiveness and take corrective actions where responses do not match expectations.
- **Communicate:** it is a step that is carried out continually. It ensures that the threats and opportunities faced by the project are communicated both internally within the project and externally with other stakeholders. Risks are communicated as part of the period project reports.

The project board is responsible for all aspects of risk management. The project manager is responsible for maintaining the risk register along with the support from the project support team. In addition, the project manager is responsible for ensuring that risks are being identified, assessed and controlled throughout the project lifecycle.

Figure 4 Risk management procedure



There will be a risk register that will be maintained throughout the project using Excel in support of the risk management procedure. The purpose of the Risk Register is to capture and maintain information on all of the identified threats and opportunities relating to the project. Each risk on the Risk Register is allocated a unique identifier as well as details such as: who raised the risk, when it was raised, the category of risk, probability, impact, response, risk status and risk owner. Project support team will help the project manager in maintaining the risk register.

The following are some of the possible risks that are expected in the development and implementation of the project.

Table 5 Risk Matrix

Risk	The degree of risk probability			Solution
	Low	Middle	High	
Limited knowledge of End Users				By working together, the client identifies knowledge gaps and consistently provides the necessary training.
Original requirements might be changed or new requirement could be added outside of scope				Documents regularly review and work boundaries for any changes in boundaries. It will be handle by change management
Sickness and resignations will reduce the number of developers available to work on this project				Assign developer/employee immediately
ONRSLB responsible body may not describe their requirement clearly				The two management will working together to solve problems

20. Project Product Description and Deliverables

In the completion of this project, INSA will submit the following documents to the OLUP according to the project plan. The following table contains a list of deliverable documents.

Table 6 Project planning and monitoring system deliverables

No	Product (Artifact)	Format
1	BRS, SRS and SAD document	Hard Copy
2	System source code	Soft Copy
3	Deployment ,Installation and Configuration document	Hard Copy
4	User manual	Hard Copy
5	Master test plan	Hard Copy
6	Test case	Hard Copy
7	Test Plan	Hard Copy

21. Project Phasing and Tentative Timeline

The time required to accomplish land use planning and monitoring system, which sums up to give the total project's development lifetime is given below.

Note: The number of days mentioned in the table are considering only national working days.

Table 7 Project Phasing and Tentative Timeline

No.	Work package description	Deadline	Remarks
1	Technical and Financial Proposal	Nov 5, 2022	
3	Baseline Assessment and requirement specification (SRS) Report	Nov 20-2022	1 week after the contractual agreement
4	Progress report	Every other month	
5	System Requirement Specification	Dec 5 ,2022	2 weeks after baseline assessment report.

	(SRS) document		
6	First stage stakeholders workshop	Dec 7, 2022	After the finalization of the SRS document
8	System architecture design document /SAD/	Jan 9,2023	1 months after SRS delivery
9	Second stage stakeholders workshop	Jan 11,2023	After accomplishment of SAD
10	The first phase of a full-fledged system	3 months after Second stage stakeholders workshop	• DMS Integration • RRR and Party registration • Subsequent registration • Deployment on pilot sites • Testing
11	First phase end-users / system administrator's training and testing	1 week	After the first phase developed system finalization
12	Second phase full-fledged system	2 months	• Online service delivery • Payment integration • Land bank • Deployment on pilot sites • Testing
13	Second phase end-users / system administrator's training and testing	1 week	• After the second phase developed system finalization
14	Third phase full-fledged system	2 weeks	• Finalize the system based on phase one and phase two test findings • Expanding the coverage area accordingly (N.B: - Based on further discussion) • Acceptance test

16	Other deliverables mentioned in section 20	1 week	On the due process of the above signs of progress
Total		8 months	

22. Warranty services

Following the acceptance of the system INSA will start the warranty services that will be provided for duration of 6 months. The warranty services will overlap with the operational support activity initially.

The following are the services that will be provided during the warranty period.

- Providing remote support on the use and administration of the system
- Correct errors in the system
- Modifying the system to include new features

In order to provide the above services INSA will maintain the following infrastructure at its premises

- A complete installation of the City, Sub-City and Kebele and region level ILUPM system
- All the development and testing platforms used during the system development
- Developers responsible for each component of the system
- A warranty service manager
- Issue tracking system
- Email and phone communication infrastructure

For the warranty service period the client will be requested to maintain the following infrastructure

- First level support (help desk personnel) at each City, Sub-City and Kebele s: the first level support team will be the main contact point of the client to receive issues. If the team will classify the issue and tries to resolve it. If the issue is not resolved it will escalate it to the second level support.

- Second level support at center: the second level support is a team that will possibly reside inside the federal urban real property registration and information agency. The second level support will receive issues from the first level support, classify the issues and tries to solve them. If the issue is not resolved it will escalate it to third level support (INSA).

The following procedure will be followed during the warranty period.

- End user: end users will report an issue to the help desk officers.
- Help desk: registers the issue and tries to solve it. If the issue is not resolved it will be passed on to the second level support.
- Second level support: second level support will receive the escalated issues, checks its type and severity classification and tries to solve it. If the issue is not resolved it will be passed on to the third level support (INSA).
- Third level support: this is a back office support team located at INSA, in Addis Ababa. The issue classification will be checked; if it is an error it will resolve it; but if it is a change request a separate proposal that describes the duration it will take to implement, how the solution will be delivered and installed, the cost for implementing the request will be submitted. Once an agreement on the proposal is acquired from the client INSA will implement the change. There are two possible ways to handle financial aspects for change request. One is to include in the warranty agreement the cost for a defined effort (hours) and to deduct from the provided hours for each change request till they are exhausted. The other option is to bill each change request separately. The choice on how to handle the financial aspect will be decided by the client when the warranty proposal is submitted at the end of the project.

Issue Types

Table 8 Issue Type

Type	Description
Error	The issue is an error in the system and it does not match to the original specification
Works as designed	The system works according to the specification
Usage error	The issue is a result of wrong usage
Change request	The issue is not an error, but a new requirement which was not part of the

	original specification. Nonetheless it is an issue that restricts the usability of the system..
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Issues will be classified according to their severity as follows

Table 9 Severity level of issue

Type	Description
Critical	Issue blocks the usage of the entire system or major components
Major	Issue has a significant impact the system but does not block the usage of the system.
Minor	The issue will not prevent the user fulfilling his/her tasks.

Handling of issue

Table 10 Handling of issues

Type	Description
Error	Issue blocks the usage of the entire system or major components
Works as designed	Issue has a significant impact the system but does not block the usage of the system.
Usage error	The issue will not prevent the user fulfilling his/her tasks.
Change request	For each change request the client need to provide a specification. Based on the specification INSA will submit a proposal for each change request that describes the duration it will take to implement, how the solution will be delivered and installed, the cost for implementing the request will be submitted. Once an agreement on the proposal is acquired from the client INSA will implement the change. There are two possible ways to handle financial aspects for change request. One is to include in the warranty agreement the cost for a defined effort(hours) and to deduct from the provided hours for each change request till they are exhausted. The other option is to bill each change request

	separately. The choice on how to handle the financial aspect will be decided by the client when the warranty proposal is submitted at the end of the project.
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The response time and type for errors will vary according to their severity

Table 11 Issue response time and type

Severity	Response Time	Response Type
Critical	2 working days	Hot patch: Will be delivered immediately after rectifying the error
Major	5 working days	Patch. Will be part of the next patch set. Delivery as agreed.
Minor	10 working days	Next version of the software. Delivery as agreed

23. Post warranty services

After the end of the warranty services INSA will provide a post warranty service for a period of 6 months' years. The post warranty services will follow the same procedure to the warranty services. But additional services could be required or existing ones be reduced. A detail post warranty service proposal along with the warranty service proposal will be submitted to the client at the end of the project.